

Sharp gaps for positive multilinear relaxations

Abstract

Luedtke, Namazifar, and Linderoth conjectured that, for positive-coefficient multilinear functions on nonnegative boxes, the term-by-term relaxation gap is within a universal constant of the convex-hull gap. We give a sparse unit-coefficient family that disproves this conjecture. Its prescribed point is strictly interior, its termwise gap is L , and its exact hull gap is $s + (L - s)2^{-s}$ for an explicitly specified cutoff s . A finite inequality and an attaining probability law prove the formula. We also determine the sharp leading growth of the worst ratio: $R(d) \sim \ln d / \ln \ln d$ in maximum degree and $C(n) \sim \ln n / \ln \ln n$ in ambient dimension. The universal upper bound uses one distribution for all terms and extends to the original termwise relaxation on every finite nonnegative box. The accompanying standalone Lean project verifies the disproof, exact hull-gap formula, and both leading-constant-one limits. A claim-to-declaration guide and reproduction instructions accompany the proofs.

1 The conjecture and the gap being compared

Let $B = \prod_{i=1}^n [\ell_i, u_i]$ be a finite box with $0 \leq \ell_i \leq u_i$. Consider a specified multilinear polynomial

$$f(x) = \sum_{e \in E} a_e m_e(x), \quad m_e(x) = \prod_{i \in e} x_i, \quad a_e \geq 0, \quad (1)$$

where E is a set of distinct subsets of $[n]$. Zero coefficients can be deleted; thus the positive-coefficient class is included. Empty supports and singleton supports are allowed and contribute only affine terms. For a continuous function g on B , write $\text{vex}_B g$ for its greatest convex underestimator and $\text{cav}_B g$ for its least concave overestimator. Define

$$H_B(f; x) = \text{cav}_B f(x) - \text{vex}_B f(x), \quad (2)$$

$$T_B(f; x) = \sum_{e \in E} [\text{cav}_B(a_e m_e)(x) - \text{vex}_B(a_e m_e)(x)]. \quad (3)$$

The first quantity is the vertical width of the convex hull of the graph of f . The second gives each original monomial its exact graph hull separately. Always $0 \leq H_B \leq T_B$. We form T_B/H_B only when $H_B > 0$.

Conjecture 4.1 of Luedtke et al. [2, p. 349] proposes a constant upper bound on this ratio, uniform over positive multilinear functions on nonnegative boxes. The same statement is Conjecture 1 on printed page 22 of the linked author manuscript. The uniformity includes varying degree and dimension. The construction below stays on unit cubes and uses coefficient one for every included term.

The source distinguishes exact termwise envelopes from recursive McCormick relaxations of higher-degree products on general boxes. Our upper bounds concern (3). They do not assert the same bounds for every recursive formulation on a positive box. The mixed-sign bilinear gap results of Boland et al. [1] address a different coefficient class. A pointwise width ratio also does not by itself

give an approximation ratio for a constrained optimization problem or an algorithm for evaluating a convex envelope.

For $d \geq 2$, let $R(d)$ be the supremum of T_B/H_B over (1) of maximum degree at most d , arbitrary finite dimension, all such boxes, and all points with positive hull gap. Let $C(n)$ be the analogous supremum over exactly n ambient coordinates, with no separate degree restriction. Unused coordinates and boundary points are permitted. Fixed coordinates ($\ell_i = u_i$) are permitted in both definitions. Constants can be handled without division by a zero gap.

The paper proceeds from an elementary disproof to the exact finite gap and then to the sharp replacement. Appendix A gives the complete common-coupling upper argument. Appendix B connects the statements to the bundled Lean declarations. All mathematical ingredients needed for the argument are proved below; the bibliography identifies the conjecture and its immediate context.

2 Graph hulls and probability laws

We omit the subscript B on a unit cube. A polynomial is multilinear when it is affine in each coordinate separately.

Lemma 1 (Vertex laws). *For a multilinear g on a finite box B , let $\mathcal{P}_B(x)$ be the probability laws on its vertices having mean x . Then*

$$\text{vex}_B g(x) = \min_{P \in \mathcal{P}_B(x)} \mathbb{E}_P g(X), \quad \text{cav}_B g(x) = \max_{P \in \mathcal{P}_B(x)} \mathbb{E}_P g(X).$$

Both extrema are attained, and these are the endpoints of the vertical slice of the original continuous graph hull.

Proof. Round each nonfixed coordinate of a point $z \in B$ independently to its two endpoints, preserving its mean. Repeated use of separate affinity gives $\mathbb{E}X = z$ and $\mathbb{E}g(X) = g(z)$. A fixed coordinate stays fixed. Thus every continuous graph point lies in the convex hull of the vertex graph. The reverse hull inclusion is immediate. A convex combination of vertex graph points is exactly a vertex law with its mean and expected function value. Laws with a specified mean form a nonempty compact polytope; hence minimizing and maximizing the expected value attain the two endpoints. $\square$

Lemma 2 (Individual envelopes and a common upper law). *For a nonempty support e on the unit cube,*

$$\text{cav } m_e(x) = \min_{i \in e} x_i, \quad \text{vex } m_e(x) = \max \left(0, \sum_{i \in e} x_i - |e| + 1 \right). \quad (4)$$

A single law attains the upper envelope of every monomial simultaneously. Consequently, $\text{cav } f(x) = \sum_e a_e \text{cav } m_e(x)$ when $a_e \geq 0$.

Proof. On a binary vertex, a monomial indicates simultaneous success of all its coordinates. The joint-success probability is at most the smallest success marginal and at least one minus the sum of the failure marginals, or zero if that is larger. For the upper bound, take one uniform $U \in [0, 1]$ and set $X_i = \mathbf{1}\{U \leq x_i\}$ for every coordinate. This attains all upper bounds at once.

For the lower bound for one support, place intervals of lengths $1 - x_i$ consecutively on a circle of circumference one. Their union has length $\min(1, \sum_{i \in e} (1 - x_i))$: before one full turn the intervals are disjoint up to endpoints, and after a full turn they cover the circle. Let each interval be the corresponding failure event. This realizes the claimed joint-success probability. Sample unused coordinates with their prescribed means to complete the law. Nonnegative coefficients allow the common upper equalities to be summed. A constant term is immediate. $\square$

These elementary envelope facts are also part of the classical multilinear relaxation theory discussed in Luedtke et al. [2]. The essential distinction is that each term can have its own minimizing law in T , while the convex envelope of the sum requires one law for all terms together.

3 A sparse family with unbounded ratio

Fix an integer $L \geq 2$ and put $m = 2^L$. Use leaves $z_0, \dots, z_{m-1}$ and anchors $a_1, \dots, a_L$. At level j , partition the leaf indices into consecutive blocks

$$\mathcal{B}_j = \{\{b2^{L-j}, \dots, (b+1)2^{L-j} - 1\} : 0 \leq b < 2^j\}.$$

The partitions are nested. Define the polynomial and evaluation point by

$$f_L(a, z) = \sum_{j=1}^L \sum_{B \in \mathcal{B}_j} a_j \prod_{i \in B} z_i, \quad a_j = 2^{-j}, \quad z_i = 1 - 2^{-L}. \quad (5)$$

We write T_L and H_L for its two gaps at this point. Each support is distinguished by its anchor and its block, so all coefficients are one on distinct squarefree supports. The number of variables, number of monomials, maximum degree, and number of variable occurrences are

$$n_L = 2^L + L, \quad |E_L| = 2^{L+1} - 2, \quad d_L = 2^{L-1} + 1, \quad \sum_{e \in E_L} |e| = L2^L + 2^{L+1} - 2. \quad (6)$$

The counts follow by summing 2^j terms of size $2^{L-j} + 1$ at each level. Every prescribed coordinate lies strictly between zero and one.

Lemma 3 (Termwise gap and failure representation). *For the family (5), $T_L = L$, the upper envelope is L , and $H_L > 0$. Under any binary law with the prescribed means, let A_j be the anchor, R the number of failed leaves, and N_j the number of level- j blocks containing a failed leaf. Then*

$$\mathbb{E}R = 1, \quad \mathbb{E}A_j = 2^{-j}, \quad H_L = \max_P \mathbb{E}_P \sum_{j=1}^L A_j N_j, \quad N_j \leq \min(2^j, R). \quad (7)$$

The maximum ranges over all binary laws with all original coordinate means, without independence assumptions.

Proof. At level j , a support has $k_j = 2^{L-j}$ leaves and one anchor. Its lower envelope is $\max(0, 2^{-j} - k_j 2^{-L}) = 0$, and its upper envelope is 2^{-j} . There are 2^j such terms. Lemma 2 therefore gives $T_L = \text{cav } f_L = L$. The value at the original point is

$$f_L(a, z) = \sum_{j=1}^L (1 - 2^{-L})^{2^{L-j}} < L.$$

Since $\text{vex } f_L \leq f_L$, the hull gap is positive.

The expected number of failures is $m2^{-L} = 1$. At every binary vertex, $f_L = \sum_j A_j(2^j - N_j)$. Taking expectations gives $\mathbb{E}f_L = L - \mathbb{E} \sum_j A_j N_j$. Apply Lemma 1 to minimize $\mathbb{E}f_L$. Finally, the blocks of each partition are disjoint, so each hit block needs a different failed leaf. $\square$

The next bound applies directly to every law in (7). It requires only a finite sum.

Lemma 4 (A finite cap certificate). *For $r \geq 0$, $A_j \in \{0, 1\}$, and integers $0 \leq s \leq L$,*

$$\sum_{j=1}^L A_j \min(2^j, r) \leq sr + \sum_{j=s+1}^L A_j 2^{j-s}. \quad (8)$$

Consequently,

$$H_L \leq s + (L - s)2^{-s} \quad (0 \leq s \leq L). \quad (9)$$

Proof. Put $c_j = \min(2^j, r)$. For $j > s$,

$$A_j c_j \leq c_j - c_{j-s} + A_j 2^{j-s}.$$

If $A_j = 0$ this uses $c_j \geq c_{j-s}$; if $A_j = 1$ it uses $c_{j-s} \leq 2^{j-s}$. Bound the first s selected caps by c_j and sum. The cap differences telescope to $\sum_{j=L-s+1}^L c_j \leq sr$, proving (8). Now use $N_j \leq \min(2^j, R)$ and take expectations. Each term in the last sum has expectation $2^{j-s}2^{-j} = 2^{-s}$, and $\mathbb{E}R = 1$. $\square$

Theorem 5 (Disproof of the uniform-constant conjecture). *For every real K there is a unit-coefficient multilinear polynomial on a unit cube, at a strictly interior point with positive hull gap, for which $T/H > K$. In particular, the Luedtke–Namazifar–Linderoth uniform-constant conjecture is false.*

Proof. Choose $s = \lceil \log_2 L \rceil$ in (9); then $s \leq L$ and $(L - s)2^{-s} \leq 1$. Hence

$$\frac{T_L}{H_L} \geq \frac{L}{\lceil \log_2 L \rceil + 1} \rightarrow \infty.$$

All coefficient, support, interiority, and positive-denominator conditions were checked in (5) and Lemma 3. A unit cube is an admissible nonnegative box. $\square$

The degree grows with L . This theorem is consistent with finite degree-dependent bounds, whose sharp growth is determined below. A single example exceeding a particular numerical ratio would not suffice to disprove the existence of every universal constant; the family does.

4 The exact hull gap and an attaining law

The finite bound is attained at a suitable cutoff. Define

$$B_q = (L - q + 2)2^{-q} \quad (1 \leq q \leq L).$$

This sequence decreases strictly, with $B_1 \geq 1 \geq B_L$. There is therefore an integer $1 \leq s < L$ such that

$$B_{s+1} \leq 1 \leq B_s, \quad \text{or equivalently} \quad (L - s + 1)2^{-(s+1)} \leq 1 \leq (L - s + 2)2^{-s}. \quad (10)$$

Theorem 6 (Exact finite gap). *For every $L \geq 2$ and every cutoff s satisfying (10),*

$$T_L = L, \quad H_L = s + (L - s)2^{-s}. \quad (11)$$

Both envelope endpoints are attained. Either valid adjacent cutoff at an equality in (10) gives the same value.

Proof. The upper bound for H_L is Lemma 4. To attain it, first choose $l \in \{0, \dots, L\}$ with probabilities

$$w_l = \begin{cases} 2^{-l-1}, & l < L, \\ 2^{-L}, & l = L. \end{cases} \quad (12)$$

Set $A_j = \mathbf{1}\{j \leq l\}$. Summing the geometric tail gives $\mathbb{P}(A_j = 1) = 2^{-j}$. For a cutoff $q \in \{s, s+1\}$, prescribe

$$R_l^{(q)} = \begin{cases} 0, & l < q, \\ 2^{l-q+1}, & l \geq q. \end{cases} \quad (13)$$

The failure count is at most 2^L , since $q \geq 1$. A direct sum gives

$$\mathbb{E}R^{(q)} = (L - q)2^{-q} + 2^{1-q} = B_q.$$

Independently of l , mix the two choices of q with weights

$$\theta = \frac{1 - B_{s+1}}{B_s - B_{s+1}}, \quad 1 - \theta = \frac{B_s - 1}{B_s - B_{s+1}}. \quad (14)$$

They lie in $[0, 1]$ and make the mixed failure count have mean one.

We next realize the block hits and each individual leaf mean. If the chosen failure count is zero, let every leaf succeed. If it is $R = 2^r$, choose a uniform residue t modulo 2^{L-r} and fail exactly the leaves whose indices are congruent to t modulo 2^{L-r} . This gives 2^r failed leaves and conditional failure probability $R/2^L$ for every leaf. At level j , a block has length 2^{L-j} . If $j \leq r$, each block contains a failure, so the number hit is 2^j . If $j > r$, the spacing between failures exceeds the block length, and the 2^r failures hit different blocks. In both cases,

$$N_j = \min(2^j, R) \quad (15)$$

simultaneously for all levels. Averaging $R/2^L$ over the mixed law gives failure probability 2^{-L} for each leaf. Thus the entire law has the original prescribed means.

It remains to check the attained objective. For the nested anchors put $S_l(R) = \sum_{j=1}^l \min(2^j, R)$. For $l > s$ and either $R = 2^{l-s}$ or $R = 2^{l-s+1}$,

$$S_l(R) = sR + 2^{l-s+1} - 2 = sR + \sum_{j=s+1}^l 2^{j-s}.$$

Indeed, cap the first $l-s$ terms at 2^j and the last s at R ; the two endpoint choices both give equality. These are exactly the failure counts in (13). For $l = s$ they are 0 and 2, both satisfying $S_s(R) = sR$; for $l < s$ they are both zero. Therefore (8) is an equality in every state of the mixed law. Taking expectations and using (15) gives $\mathbb{E} \sum_j A_j N_j = s + (L-s)2^{-s}$. This attains the maximum in (7), and hence attains the lower envelope $L - H_L$. The common upper law attains L .

Finally, the formulas at s and $s+1$ differ by $B_{s+1} - 1$, so they agree whenever both adjacent choices are valid. $\square$

The cutoff also explains the growth. From $2^s \leq L - s + 2 \leq L + 1$, $s \leq \log_2(L + 1)$. The other cutoff inequality gives $2^{s+1} \geq L - s + 1$, which, together with this upper bound, yields $s = \log_2 L + O(1)$. Moreover,

$$0 \leq (L - s)2^{-s} \leq 2 \frac{L - s}{L - s + 1} < 2.$$

L	A valid s	H_L	T_L/H_L
2	1	3/2	4/3
8	2	7/2	16/7
16	3	37/8	128/37
64	5	219/32	2048/219

Table 1: Exact instances of Theorem 6. These values illustrate the formula; the universally quantified proof establishes unboundedness.

Consequently,

$$H_L = \log_2 L + O(1), \quad \frac{T_L}{H_L} \sim \frac{L}{\log_2 L}. \quad (16)$$

Nesting of the partitions was used for simultaneous attainment in (15). The finite bound and disproof themselves only need the specified equal block sizes and disjointness within each level.

5 The sharp replacement

The construction gives the sharp leading scale for both degree and dimension. Here and below $\ln$ denotes the natural logarithm.

Theorem 7 (Sharp degree and dimension growth). *For the suprema defined in Section 1,*

$$R(d) \sim \frac{\ln d}{\ln \ln d}, \quad C(n) \sim \frac{\ln n}{\ln \ln n}. \quad (17)$$

Both limits hold along all positive integer allowances tending to infinity, with leading constant one, over all finite nonnegative boxes.

We state the explicit bound used for the upper half. It matches the finite harmonic certificate in the accompanying Lean proof.

Theorem 8 (A common-coupling finite bound). *Let $d \geq 2$, $c_0 = 1 - e^{-1}$, and define*

$$\Lambda = \max\{e^6, 1 + \ln d\}, \quad b = \ln \Lambda, \quad A = \Lambda/b^2, \quad (18)$$

$$h = \frac{1 - 1/b}{1 + 1/b^2} \frac{b - 3 \ln b}{\Lambda}, \quad Z = \frac{1}{h} + A + \frac{2}{c_0}. \quad (19)$$

Then $h > 0$ and $T_B(f; x) \leq Z H_B(f; x)$ for every positive multilinear polynomial of degree at most d on a finite nonnegative box, including points of zero hull gap. On a unit cube the bound is witnessed by a single law depending only on the complete marginal vector and d , independently of the supports and coefficients.

Appendix A proves this theorem. The finite constant is chosen to give the sharp leading term; no optimality of Z at a fixed degree is asserted.

Proof of Theorem 7. For the upper bound, $b \rightarrow \infty$ and

$$\frac{h}{b/\Lambda} = \frac{1 - 1/b}{1 + 1/b^2} \left(1 - \frac{3 \ln b}{b}\right) \rightarrow 1.$$

Thus $1/h \sim \Lambda/\ln \Lambda$, while $A/(\Lambda/\ln \Lambda) = 1/b \rightarrow 0$ and the constant $2/c_0$ is negligible. Eventually $\Lambda = 1 + \ln d$, so Theorem 8 gives

$$R(d) \leq (1 + o(1)) \frac{\ln d}{\ln \ln d}.$$

Every support in n coordinates has size at most n , so $C(n) \leq R(n)$.

For a common lower argument, take an integer $N \geq 8$ and $L = \lfloor \log_2 N \rfloor - 1$. Then $L \geq 2$ and $n_L = 2^L + L \leq 2^{L+1} \leq N$. The family therefore fits degree allowance N . It also fits exactly N coordinates after adding unused variables with arbitrary fixed cube means, for instance $1/2$. Projection of any law on the added coordinates gives an old law, and any old law extends by independent added coordinates. Both gaps are therefore unchanged. By (16), both $R(N)$ and $C(N)$ are at least

$$\frac{T_L}{H_L} \sim \frac{L}{\log_2 L} \sim \frac{\ln N}{\ln \ln N}.$$

This construction works for every $N \geq 8$. Together with the upper bounds it proves both limits. In particular, the relevant ratio sets are nonempty and bounded above before taking their suprema. $\square$

The conjectured constant is therefore replaced by a slowly growing degree- or dimension-dependent factor. The lower family has fewer than $2n_L$ terms and $O(n_L \log n_L)$ variable occurrences. These elementary size consequences do not claim an efficient algorithm for constructing or evaluating a general convex envelope.

A The common-coupling upper bound

We give all details needed for Theorem 8. The parameters and the rational union estimate below follow the formal proof, so no optimized fixed-point or second-order certificate is needed.

A.1 Deficiencies and the easy terms

Discard constant and affine terms, which have zero width and do not change H . For a remaining support e , choose an anchor i_e of minimum mean and write

$$u_e = x_{i_e}, \quad p_j = 1 - x_j \ (j \neq i_e), \quad t_e = \min \left(u_e, \sum_{j \in e \setminus \{i_e\}} p_j \right), \quad D_e(X) = X_{i_e} - \prod_{i \in e} X_i. \quad (20)$$

For binary X , D_e is the nonnegative indicator that the anchor succeeds and at least one other coordinate fails. Lemmas 1 and 2 imply

$$T(f; x) = \sum_e a_e t_e, \quad H(f; x) = \max_{P \in \mathcal{P}(x)} \sum_e a_e \mathbb{E}_P D_e. \quad (21)$$

Thus a law satisfying $\mathbb{E} D_e \geq t_e/Z$ for every support bounds the ratio after summing with nonnegative coefficients. The law must be chosen once for the entire vector x .

Call a coordinate low if $x_i \leq 1/2$ and high otherwise. Independent Bernoulli rounding gives

$$\mathbb{E} D_e = u_e \left(1 - \prod_{j \in e \setminus \{i_e\}} (1 - p_j) \right).$$

If there are at least two low coordinates in the support, a nonanchor has $p_j \geq 1/2$, so the deficiency is at least $u_e/2 \geq t_e/2$. If there are no low coordinates, let $S = \sum_j p_j$. The elementary inequality $1 - p \leq e^{-p}$ gives

$$\mathbb{E}D_e \geq u_e(1 - e^{-S}) \geq c_0 u_e \min(1, S) \geq c_0 t_e/2.$$

For the middle inequality, use concavity of $1 - e^{-S}$ on $[0, 1]$ and monotonicity after 1. For the last, if $S \leq 1$ use $u_e > 1/2$; otherwise use $t_e \leq u_e$. Independence therefore captures at least $c_0 t_e/2$ in these two cases. Terms of zero gap need no division and satisfy the desired inequality under every law.

A.2 Two laws for the one-low case

Put $M = e^{\Lambda-1} \geq d$, so $1 + \ln M = \Lambda$. Each of the next two laws draws one uniform $U \in (0, 1)$ and sets every low coordinate to $X_i = \mathbf{1}\{U \leq x_i\}$.

In the threshold law, a high coordinate with failure mean $p = 1 - x_i$ fails exactly when $U \leq p$. All means are preserved. For a one-low support with low-anchor mean u , this law gives deficiency

$$\min\left(u, \max_j p_j\right). \quad (22)$$

In the harmonic law, a high coordinate with $p = 0$ never fails. For $p > 0$, put

$$h_p = \min(Mp, 1), \quad a_p = 1 + \ln(h_p/p), \quad q_p(t) = \frac{\min(1, p/t)}{a_p} \mathbf{1}\{t \leq h_p\} \quad (0 < t < 1). \quad (23)$$

Conditional on $U = t$, high coordinates fail independently with probabilities $q_p(t)$. Since $p \leq h_p$ and $1 \leq a_p \leq \Lambda$, these probabilities belong to $[0, 1]$. They are measurable and

$$\int_0^1 q_p(t) dt = \frac{p + p \ln(h_p/p)}{a_p} = p. \quad (24)$$

Hence every high coordinate also has the prescribed mean. Values at the endpoints of the integral have no effect. The finite binary law can equivalently be defined by integrating the conditional probability of each binary vertex; finite sums commute with the integral.

For a one-low support its anchor is the low coordinate. Conditional on $U = t \leq u$, the probability of a nonanchor failure is $1 - \prod_j (1 - q_{p_j}(t))$. Its harmonic deficiency is therefore

$$\int_0^u \left[1 - \prod_j (1 - q_{p_j}(t)) \right] dt. \quad (25)$$

Both laws are defined using the complete marginal vector, without reference to a support or its coefficient.

A.3 The harmonic window and the mixture

First, for independent events of probabilities $q_j \in [0, 1]$ and $S = \sum_j q_j$,

$$1 - \prod_j (1 - q_j) \geq \frac{S}{1 + S}. \quad (26)$$

Indeed, $\prod_j (1 + q_j) \geq 1 + S$ and $\prod_j (1 - q_j) \prod_j (1 + q_j) = \prod_j (1 - q_j^2) \leq 1$. Multiply the first inequality by the nonnegative failure-complement product and rearrange. The right side is increasing in $S \geq 0$.

Let $\beta = 1/b$, with b, A, h as in (18)–(19). We have $b \geq 6$, and $b - 3 \ln b > 0$: it is positive at 6 (since $e^2 > 6$) and its derivative is $1 - 3/b > 0$ thereafter. Thus $h > 0$ and

$$\ln(\beta A) = b - 3 \ln b > 0, \quad \beta A > 1, \quad A > 1.$$

For a one-low term with $t_e > 0$, if some $p_j \geq t_e/A$, then (22) is at least t_e/A , since $t_e \leq u$. Otherwise every $p_j < t_e/A$. Consider the nonempty window

$$t_e/A \leq t \leq \beta t_e \leq u. \quad (27)$$

All $p_j \leq t$ there. Call a coordinate active when $t \leq M p_j$. Since $t \leq 1$, this is the same as $t \leq h_{p_j}$. Every inactive coordinate has $p_j < t/M$, and there are at most $d - 1 \leq M$ coordinates in the nonanchor support. The total inactive failure mass is at most t , so the active mass is at least

$$\sum_j p_j - t \geq t_e - t \geq (1 - \beta)t_e.$$

On active coordinates, (23) gives $q_{p_j}(t) \geq p_j/(\Lambda t)$. Consequently,

$$\sum_j q_{p_j}(t) \geq z(t) := \frac{(1 - \beta)t_e}{\Lambda t}, \quad 0 \leq z(t) \leq \frac{A}{\Lambda} = \frac{1}{b^2}.$$

Using (26), the conditional union probability is at least $z(t)/(1 + z(t)) \geq z(t)/(1 + 1/b^2)$. Integrate over (27), using nonnegativity outside that window, to obtain

$$\begin{aligned} \mathbb{E}_{\text{harm}} D_e &\geq \frac{1 - \beta}{1 + 1/b^2} \frac{t_e}{\Lambda} \ln(\beta A) \\ &= \frac{1 - 1/b}{1 + 1/b^2} \frac{b - 3 \ln b}{\Lambda} t_e = h t_e. \end{aligned}$$

The integrands are bounded measurable functions on a finite interval, and the reciprocal integration window stays strictly above zero.

Mix the harmonic, threshold, and independent laws with weights

$$\frac{1/h}{Z}, \quad \frac{A}{Z}, \quad \frac{2/c_0}{Z}, \quad Z = 1/h + A + 2/c_0. \quad (28)$$

These positive weights sum to one and preserve every coordinate mean. Each positive-gap term receives at least t_e/Z from one of the three components. All other components contribute nonnegative deficiencies. Summing (21) proves $T \leq ZH$ on the cube, including at zero hull gap. This proves the cube part of Theorem 8.

A.4 Transfer to original terms on a nonnegative box

For each coordinate substitute $x_i = \ell_i + (u_i - \ell_i)y_i$ with $0 \leq y_i \leq 1$, and let $\tilde{f}(y) = f(x)$. Expanding original monomials gives nonnegative coefficients and does not increase degree.

First suppose all coordinates are nonfixed. The substitution is an affine bijection, so it maps graph hulls and preserves their vertical widths:

$$H_B(f; x) = H(\tilde{f}; y), \quad T_B(f; x) \leq T(\tilde{f}; y). \quad (29)$$

For the second inequality, an original term becomes a sum of expanded monomials. The sum of their convex envelopes is a convex underestimator of that term; the sum of their concave envelopes

is a concave overestimator. Thus the original term’s width is at most the sum of the expanded widths. Sum this inequality over the original terms. If expansion produces the same support more than once, combining its nonnegative coefficients leaves that sum of widths unchanged, because envelopes scale linearly by a nonnegative constant.

If a coordinate is fixed, restrict the original box to its nonfixed coordinates. In the expanded function the fixed coordinate has zero width and its new cube coordinate is unused. Projecting or extending laws on unused coordinates preserves every attainable expectation, as in the proof of Theorem 7. Thus both the full-hull equality and the original-term inequality in (29) still hold. If all coordinates are fixed, every gap is zero. Applying the cube bound to $\tilde{f}$ now proves $T_B(f; x) \leq T(\tilde{f}; y) \leq ZH(\tilde{f}; y) = ZH_B(f; x)$, completing Theorem 8.

B Formal statements and reproduction

The accompanying [formal/](#) directory is a standalone Lake project pinned to Lean 4.33.1 and Mathlib v4.33.1, with exact dependency revisions in `lake-manifest.json`. No parent repository files are needed. The 48 proof modules consist of 41 multilinear modules and seven shared modules for finite laws, graph hulls, and individual envelopes. The shared namespace `CubicGap` is inherited from their original location; its bundled files are general infrastructure, and no cubic counterexample modules are included.

The following source links are relative to this PDF. They work when the PDF and the accompanying folder are kept together and the viewer permits local links. File paths and declaration names also appear in [formal/COVERAGE.md](#) for viewers that disable such links. No hosted verification service is required.

Paper claim	Lean file and principal declarations
Lemma 1, continuous cube hull	Hull.lean : <code>mem_cubeGraph_hull_iff</code> .
Lemma 3, $T_L = L$ and $H_L > 0$	Termwise.lean : <code>termwiseGap_eq</code> ; Results.lean : <code>hullGap_positive</code> .
Lemma 4	ExactUpper.lean : <code>dyadic_exact_pointwise</code> , <code>hullGap_exact_upper_bound</code> .
Theorem 5	Results.lean : <code>unbounded_gap_ratio</code> , <code>no_uniform_positive_multilinear_bound</code> .
Theorem 6	ExactResults.lean : <code>polynomial_exact_minimum</code> , <code>hullGap_exact</code> , <code>exists_hullGap_exact</code> .
Theorem 8, cube	SharpUpper.lean : <code>sharpLaw_hasMeans</code> , <code>sharp_cube_gap_bound</code> .
Theorem 8, original boxes	BoxTransfer.lean : <code>degree_gap_bound_on_box</code> .
Theorem 7	SharpAsymptotics.lean : <code>sharp_positive_growth</code> .

All multilinear declarations in the table belong to namespace `MultilinearGap`. The shared hull declarations belong to `CubicGap`. The final asymptotic theorem states the two normalized limits for the actual all-box degree and exactly- n -coordinate suprema. It is not a theorem merely along the dyadic family.

The formal disproof endpoint uses the elementary weaker estimate $H_L \leq s + 2L/2^s$ and an arithmetic parameter choice, independently of the exact formula. This paper uses the sharper finite

certificate, which is also verified, to shorten the written disproof. Similarly, the formal asymptotic lower argument uses the verified estimate $H_L \leq \log_2 L + 3$ and does not depend on the exact attaining law. The conclusions agree; these are alternative proof routes within the same bundle.

The completion modules also verify the paper’s supporting mathematical claims.

`Attainment.lean` proves general endpoint attainment, and `EnvelopeFunctions.lean` identifies those endpoints with the greatest convex underestimator and least concave overestimator on both cubes and boxes. `MonomialEnvelope.lean` proves the full individual lower-envelope identity in Lemma 2, including an attaining law preserving all ambient means.

The module `PaperFoundations.lean` checks original-box coefficient scaling and $0 \leq H_B \leq T_B$.

`ExactSize.lean` proves all four size identities in (6), with the coordinate count inherited from `FamilySize.lean`, and the stated sparsity bounds. `ExactAsymptotics.lean` proves $|H_L - \log_2 L| \leq 3$ for $L \geq 2$ and the actual family ratio limit in (16). `Examples.lean` proves all four rows of Table 1 as equalities about the actual graph hull.

With Elan, Git, and Python 3 installed and the pinned toolchain on PATH, run from `formal/`:

```
lake exe cache get
bash scripts/verify.sh
```

The first command obtains the pinned dependency cache. The script checks that every proof module is imported, builds with warnings treated as failures, audits all bundled declarations (including private helpers) for transitive axiom dependencies, and replays the compiled project declarations through Lean’s kernel. Only `propext`, `Classical.choice`, and `Quot.sound` are permitted. Unfinished proofs, custom axioms, and native-computation axioms are rejected. No external optimization solver or sampled numerical result is a trusted proof step.

The replay uses the installed Lean kernel and compiled dependencies; it is not a separate kernel implementation and does not replay all of Mathlib from scratch. The mathematical interpretation also depends on matching the formal definitions to the statements above. The [verification record](#) specifies the actual local checks and their scope. Internal agent reviews of the antecedent proofs do not constitute external peer review or establish publication priority. This note makes no exhaustive priority claim.

The formal scope is the disproof, exact family gap and attainment, the finite bound (19), and the sharp leading limits. It does not include optimized fixed-point or Lambert W formulas, second-order asymptotics, coefficient removal, or homogenization. Those results are not needed for any theorem stated here.

References

- [1] Natashia Boland, Santanu S. Dey, Thomas Kalinowski, Marco Molinaro, and Fabian Rigterink. Bounding the gap between the McCormick relaxation and the convex hull for bilinear functions. *Mathematical Programming*, 162(1–2):523–535, 2017. doi: 10.1007/s10107-016-1031-5.
- [2] James Luedtke, Mahdi Namazifar, and Jeff Linderoth. Some results on the strength of relaxations of multilinear functions. *Mathematical Programming*, 136(2):325–351, 2012. doi: 10.1007/s10107-012-0606-z. Conjecture 4.1, p. 349; Conjecture 1, p. 22 in the author manuscript, available at <https://jlinderoth.github.io/papers/Luedtke-Namazifar-Linderoth-12-TR.pdf>.